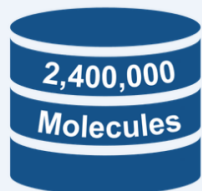


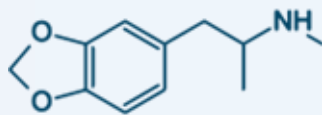
A Data

ChEMBL36

The open chemical database



Encoded as SELFIES strings.



[C] [N] [C] [Branch1] [C] [C] [C] ...

B Tokenisation

APE Tokenisation

Atom Pair Encoding

Input:

[C] [N] [C] [Branch1] [C] [C] [C] ...

APE merges

Output:

[C][N] [C][Branch1] [C][C] [C][C] ...

C Pretraining

ModernBERT

Modern transformer encoder



RoPE positional encoding



Alternating local/global attention



GeGLU feed-forward blocks



Pre-normalised transformer layers

D Evaluation

Frozen Evaluation

MoleculeNet & TDC classification tasks

MoleculeNet



Molecular Property Benchmarks



7 classification datasets



Biophysical endpoints



TDC



Therapeutics Data Commons



18 classification datasets



Toxicity, HTS & ADME endpoints

